
CEDARs were designed and built at CERN to detect secondary beam particles at the Super Proton Synchrotron (SPS). Two types of CEDAR were developed at CERN: CEDAR-W and CEDAR-N, corresponding to the areas of use. The two designs differ in terms of energy range in which they can detect beam particles, with CEDAR-N able to separate kaons and pions up to $300\text{GeV}/c$ but not detect protons below $60\text{GeV}/c$. Whereas CEDAR-W can detect protons down to $12\text{GeV}/c$ and separate kaons and pions up to $150\text{GeV}/c$.

Described in section __, NA62 has run with a CEDAR-W since 2016 filled with Nitrogen (N_2 at a pressure of 1.71 bar (absolute pressure)). In this configuration, the CEDAR contributed to the most significant amount of material in the path of the beam, introducing $39 \times 10^{-3} X_0$ of material, where X_0 is one radiation length. The aluminum windows of the CEDAR vessel contribute $3.9 \times 10^{-3} X_0$ and the N_2 gas contributes $35 \times 10^{-3} X_0$. The material from the CEDAR-W with N_2 results in multiple scattering of beam particles, which introduces an angular divergence of $32 \mu\text{rad}$. Scattered beam particles contribute to the backgrounds for the searches for ultra-rare decays, so it is imperative to reduce the amount of material in the beam to control these backgrounds. It was proposed that a CEDAR filled with Hydrogen (H_2) could reduce the material due to the radiator gas. Using H_2 as a radiator gas would require a pressure of 3.8 bar to produce a similar Cherenkov angle as N_2 at 1.71 which would instead introduce $3.4 \times 10^{-3} X_0$ of material. This would result in a 21 % reduction in total material to $7.3 \times 10^{-3} X_0$. Resulting in an angular beam divergence of $13 \mu\text{rad}$ in both the horizontal and vertical dimensions.

Taking into account that the four GTK stations contribute $20 \times 10^{-3} X_0$, the total beam material along the beam-line reduces from $59 \times 10^{-3} X_0$ to $27 \times 10^{-3} X_0$. Using

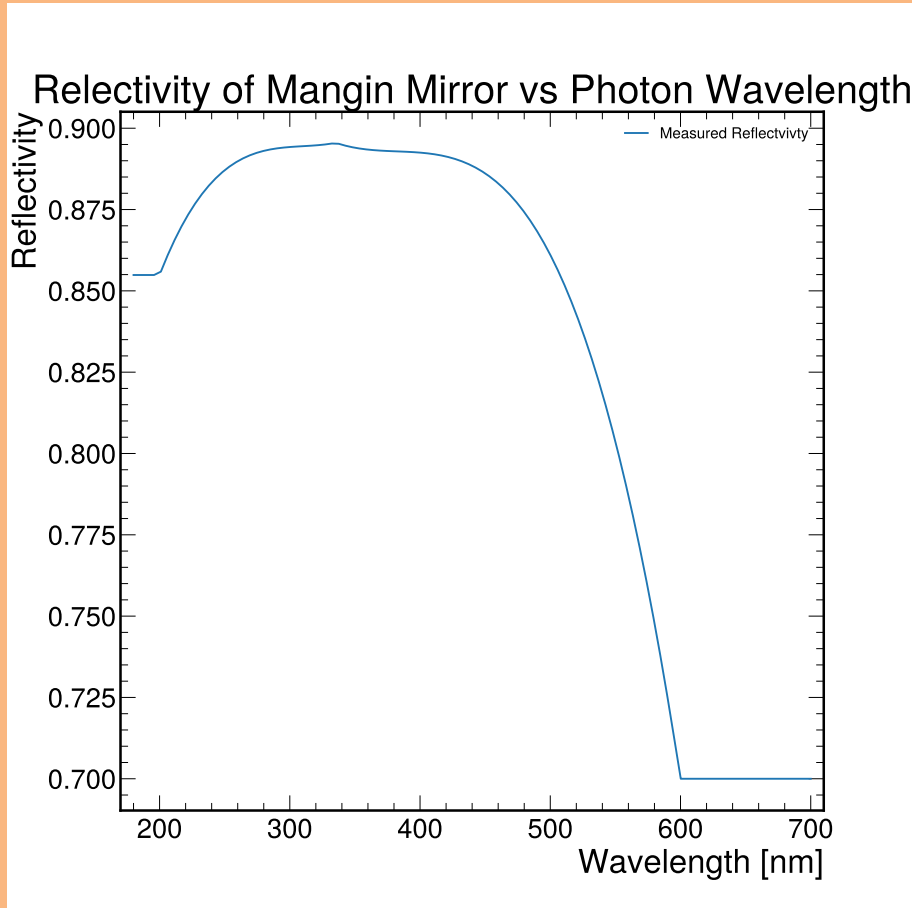


Figure 1: Measured reflectivity of CEDAR-H mangin mirror conducted by Winlight System

the full NA62 simulation based on GEANT4, it is possible to estimate that the number of particles interacting inelastically and elastically upstream of the decay volume would reduce from 2.1 % to 0.9 % (DO THIS YOUR SELF TO CHECK). This reduction in scattering would result in higher kaon transmission (number of kaons reaching the decay volume) and reduce the signal rates in downstream detectors.

Table 1: Optical and mechanical parameters of CEDAR-W and CEDAR-H. All values are dimensions in millimetres unless otherwise stated. Positions along the beam axis are quoted with respect to the upstream end of the CEDAR.

CEDAR type		CEDAR-W	CEDAR-H
Nominal gas type		N ₂	H ₂
Nominal pressure [bar]		1.71	3.80
Gas vessel cylinder	Length	4500	4500
	Inner radius	267	267
Gas vessel cap	Length	339	280
	Inner radius	139	139
Chromatic corrector	Position along the beam axis	1855	1902
	Radius of curvature	1385	1307
	Central thickness	20	20
	Inner radius	75	75
	Outer radius	160	160
Mangin mirror	Position along the beam axis	5353	5362
	Radius of curvature:		
	- refracting surface	6615	8994
	- reflecting surface	8610	9770
	Central thickness	40	40
	Inner radius	50	40
	Outer radius	150	150
Diaphragm	Position along the beam axis	872	911
	Aperture central radius	100	100
Condensers	Position along the beam axis	832	871
	Maximum thickness	10	10
	Radius of curvature	300	300
Quartz windows	Position along the beam axis	472	531
	Thickness	10	10
	Radius	22.5	22.5
	Radial distance to window centre	103	103
Optical caps	Position along the beam axis	450	450
	Maximum thickness	4.24	4.24
	Radius of curvature	114.62	114.62
Spherical mirrors	Position along the beam axis	322	322
	Radius of curvature	51.68	77.52
	Diameter	50	50
	Radial distance to mirror centre	106	106

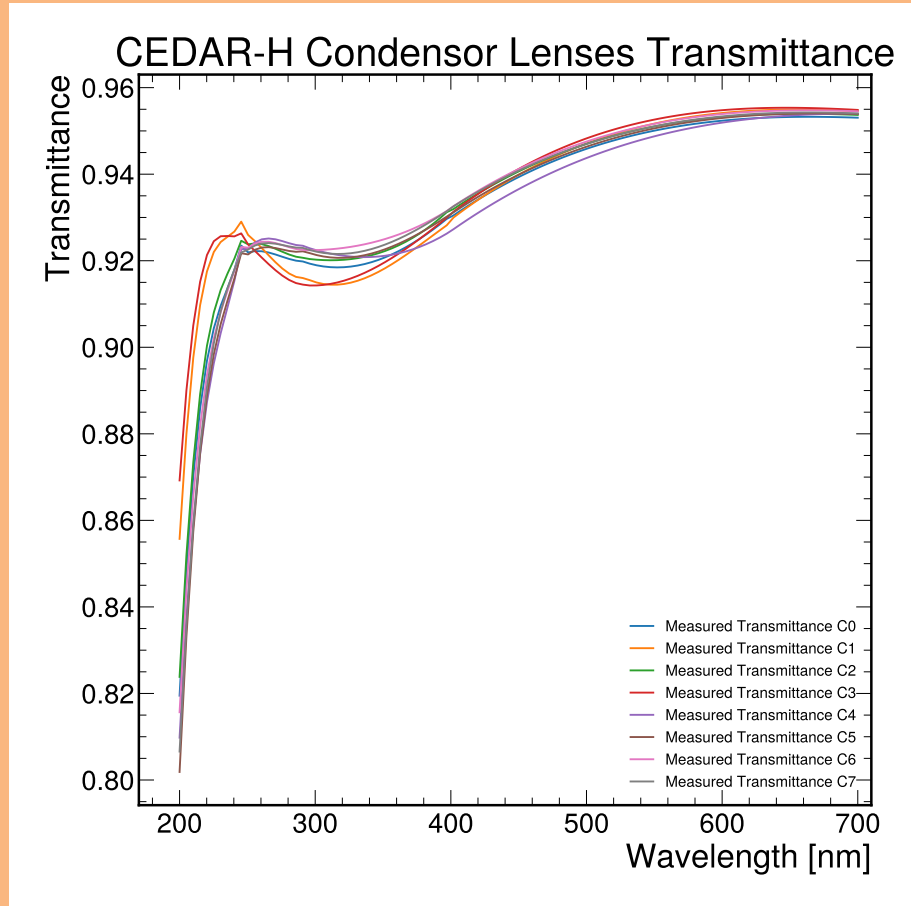


Figure 2: Measured transmittance of CEDAR-H condensor lenses conducted by Thomas Schneider at CERN

0.1 Development of CEDAR-H

0.1.1 Optical Parameters

0.2 Test-beam at CERN

0.3 Commissioning at CERN

0.4 CEDAR-H Performance at NA62

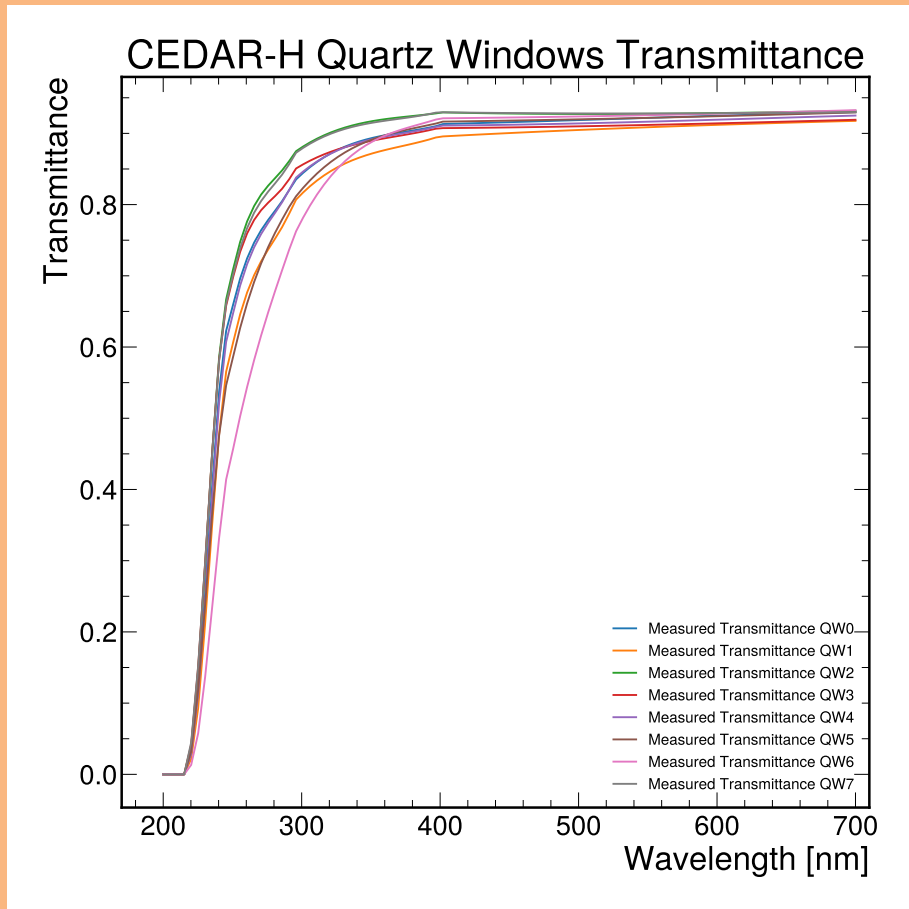


Figure 3: Measured transmittance of CEDAR-H quartz lenses conducted by Thomas Schneider at CERN

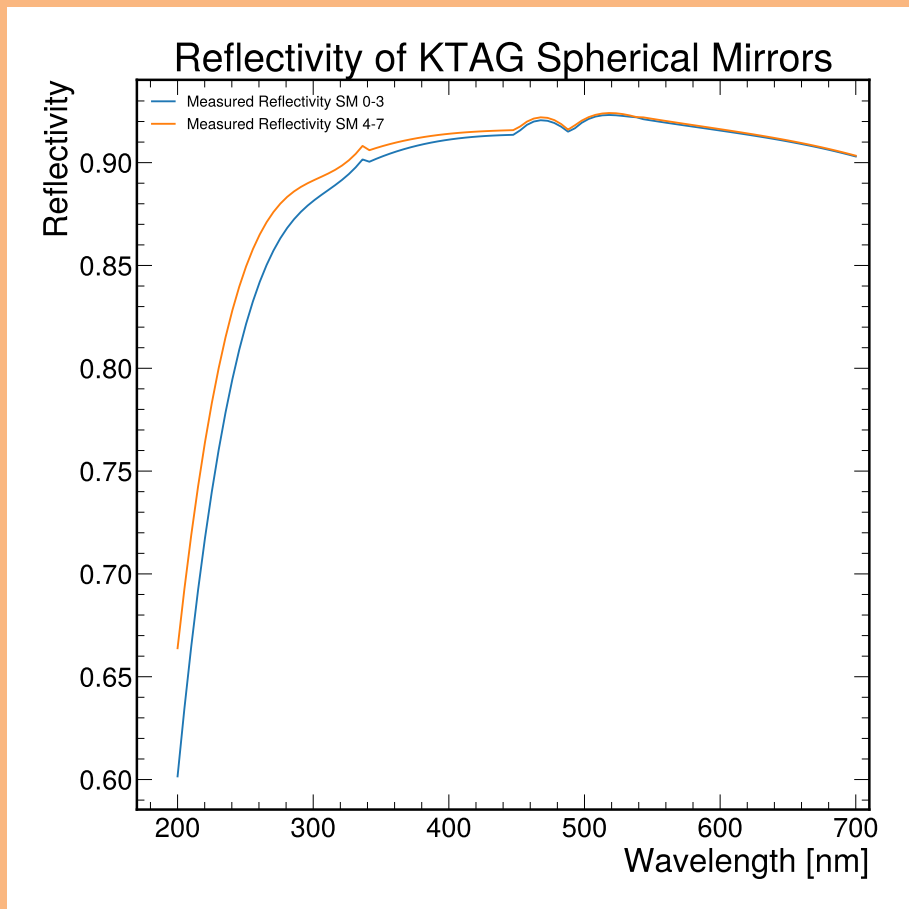


Figure 4: Measured reflectivity of KTAG spherical mirror conducted by Thomas Schneider at CERN